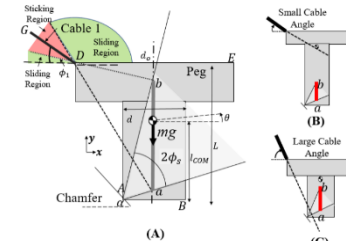
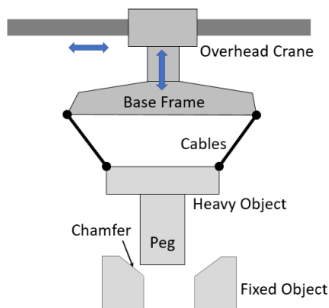
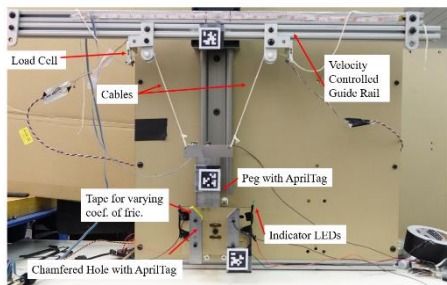


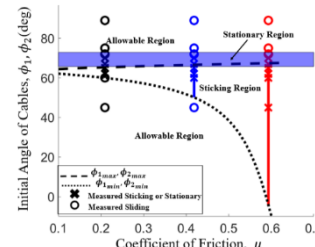
Assembly of Suspended Heavy Objects: Winchbot

Sumitomo Heavy Industries

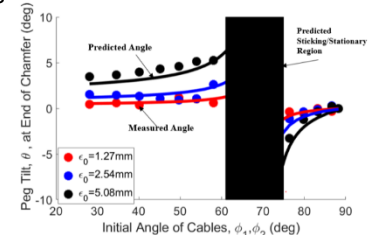
- Novel robotic solution to collaborate with workers to assembly heavy crane-hung objects
- Experimental validation proves heavy object hung with multiple cables at particular orientations can be inserted into a chamfered hole despite a small clearance
- Exploits properties of cables such that the system does not over-constrain a hung object during insertion.



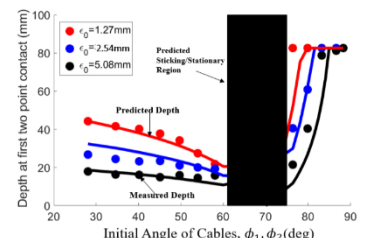
Region where the tension of the cables and the mass vector of the peg may intersect inside the friction cone during the chamfer crossing. To avoid this region, for long pegs (B), small angles are preferred and for short pegs (C), a large angle cable suspension is desired.



Experimental results for verifying No-Stick and No-Stationary region for a given peg geometry with three coefficients of friction



Predicted and measured peg tilt angle at end of chamfer crossing for the specific peg geometry in the 2D experimental setup



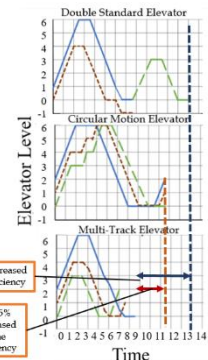
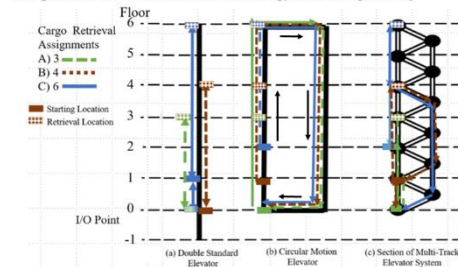
Predicted and measured depth at first two-point contact for the specific peg geometry in the 2D experimental setup

A Multi-Track Elevator System for E-Commerce Fulfillment Centers

Master's Research 2015-2017

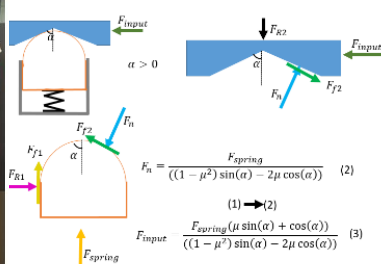
- Designed and built an efficient vertical transport system
- Developed experiments to validate efficiency of MTE system
- Coordinated MTE system control with four carriages

Objective: Determine the increased efficiency of the MTE system compared with current elevator technology based on geometry



Takeaway: The MTE architecture removes the dependence on the relative positions of the carriages, resulting in an increased time efficiency of the overall system

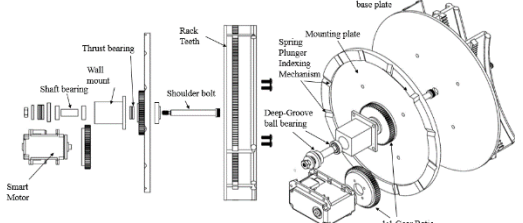
123.5% increased time efficiency



Rearranging (3) to solve for α

$$\tan(\alpha) = \frac{F_{spring} + 2\mu F_{input}}{F_{input}(1 - \mu^2) - F_{spring}\mu}$$

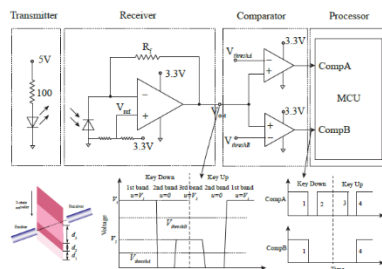
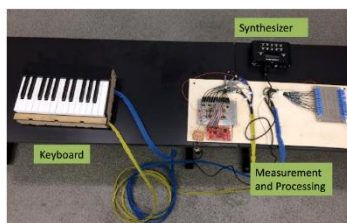
$\alpha = 123^\circ$
With chosen springs and 3Nm Motor Torque



As a student

Create a Magnetoencephalography (MEG) compatible musical instrument that enables researchers to explore neuroactivity associated with playing tonal music

- Novel 3-state optical encoder to detect key press and velocity
- Metal-free device with realistic piano feel



Medical Device Design (2.75)

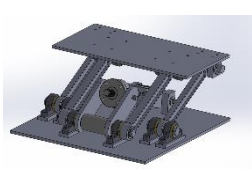
As a mentor



Lower Extremity Fracture Fixation Device



Crank Powered Otoscope



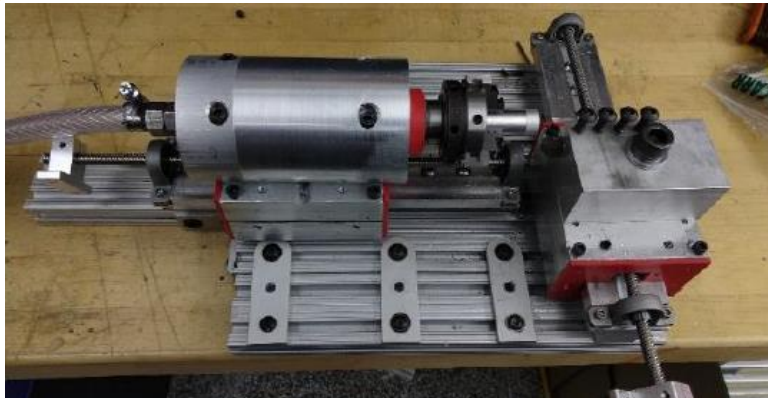
Portable Powered Lifting Device



Center of Gravity Detector for Operator Fatigue



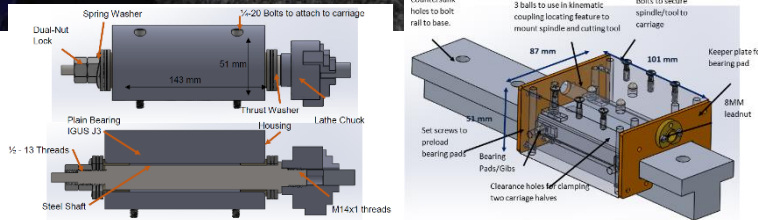
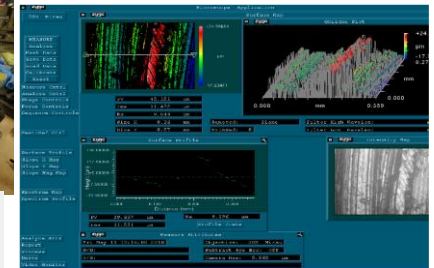
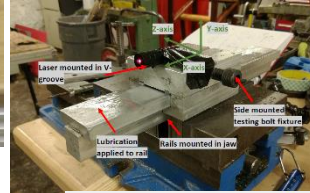
Vaginal Breech Delivery Device



Precision Machine Design (2.70) 2018

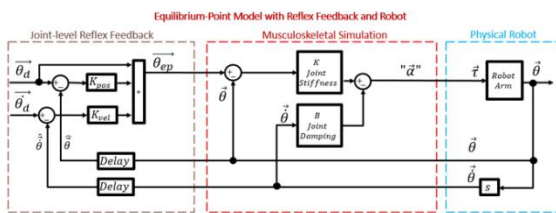
Design, Error Prediction, and Testing of a T-Based Lathe

- Measured flatness 187.9 μm $\pm$ 5.75 μm
- Error Budget Predicted flatness 233 μm $\pm$ 22.7 μm
- Custom spindle and custom boxway linear motion axes mounted perpendicular to one-another



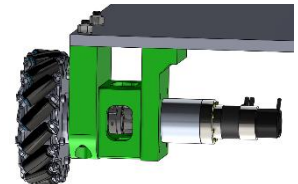
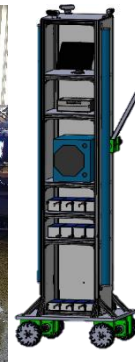
Realistic Human Reaching Movements in a Direct-Drive Robot Arm using Virtual Muscle Impedance and Equilibrium Point Control

Final Project for Biomechanics and Neural Control of Movement (2.183) 2016



Disinfection Robot (2020):

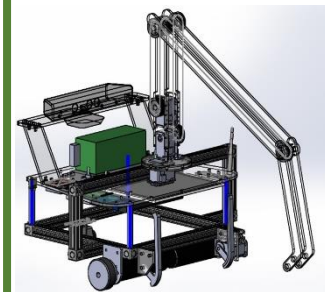
Developed a mobile platform that can be used to disinfect a room with UV light. This platform will be used for teaching introduction to robotics.



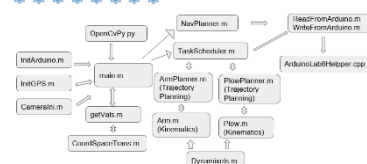
Introduction To Robotics (2.12)

Final Project 2015

- Robot designed to plow snow, use camera for navigating environment, move obstacles



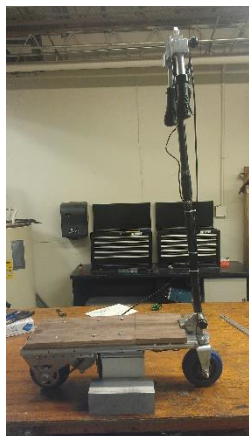
Code Architecture



Design of a Foldable Electric Scooter

Capstone Project 2015

- 15.5 mph top speed
- 220 lb weight capacity
- 10.33 ft-lb Stall Torque
- 1.5 s stop time



DrumBot

Intelligent Machine Design Lab 2015

- Translate user 'air drumming' via Xbox Kinect to remotely play the drums

